



UNIVERSITY OF
LEICESTER

**School of Computing and
Mathematical Sciences**

CO7201 Individual Project

Interim Report

AI Symptom Tracker - CogniCare

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DECLARATION

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Name: Vrushali Hippargi

Date: 26-03-2026.

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1. Project progress update

Since submitting my Preliminary Report, the development of “CogniCare” has moved from a conceptual design to a functional prototype. My primary focus has been on ensuring that the “Mind Dump” feature which is the heart of the application, is not just technically functional but clinically useful for both patients and their care circles.

1.1 Reflecting on Requirement Fulfilment:

I have evaluated my progress against the requirements; I initially set out. While most are on track, I have made several strategic pivots based on early development findings:

- **AI Engine Evolution (Essential):** Initially, I planned to use GPT-4o. However, during the implementation of the “5-Pillar” analysis, I found that **Google Gemini 1.5/2.0 Flash** offered a more expansive context window. This was a critical discovery; for a patient to see trends over months, the AI needs to “remember” a vast amount of historical data. Switching to Gemini has significantly improved the clinical accuracy of the automated summaries.
- **Building the RBAC (Role Based Access Control) (Essential):** I have successfully built the relational database using **PostgreSQL** and **Prisma**. It was important to me that this didn't feel like a standard “user” system. I've implemented a many-to-many permission structure so a patient can have many carers and a GP all in their circle, each with different levels of visibility.
- **Privacy-First AI (Essential):** One of my biggest concerns was sending sensitive health data to an external AI. To address this, I integrated a PII-scrubbing middleware (**js-pii-mask**). It now automatically redacts names and identifiers before the data leaves my server, ensuring I am building a “privacy-by-design” application.
- **The “Mind Dump” Experience:** I've integrated **Capacitor 8** to bring the app onto mobile. The most rewarding part of this phase was getting the **Voice Recorder** to work. Patients can now simply speak into their phones when they are feeling unwell, which is much more accessible than typing into a complex form.

1.2 Architectural Shifts:

The most significant change to my original plan was the decision to **decouple the backend**. I realized that a unified Next.js structure would struggle to handle the persistent connections needed for **Socket.io** (real-time carer alerts). By moving to a standalone **Express server**, the app is now much more stable and capable of handling high-performance tasks like generating complex clinical PDF reports.

1.3 Addressing Feedback from Preliminary report:

- **Refinement of AI Scope:** Initial feedback suggested ensuring the AI does not over-reach into “diagnosis” but stays within “symptom tracking”. I have adjusted the system prompts for the **Gemini 2.5 Flash** engine to focus strictly on clinical entity extraction (the 5 Pillars) instead of **OpenAI** and trend detection, explicitly adding a disclaimer that the output is for “consultation support” rather than medical advice.
- **Security & Data Sovereignty:** A key point raised was the sensitivity of patient logs. In response, I implemented the **PII-masking middleware** earlier than planned. This ensures that no identifiable data (names, specific locations) is sent to the Google Generative AI SDK, addressing the privacy concerns highlighted in the preliminary review. And instead of **Whisper**, I implemented **@yellowsakura/js-pii-mask** PII scrubbing.

- **Architecture Scalability:** Following suggestions to ensure the app handles real-time updates efficiently, I transitioned to a **Decoupled Architecture**. This allows the Socket.io server and backend to run independently of the Next.js frontend, ensuring that “Care Circle” messaging works asynchronously without affecting the performance of the main dashboard.

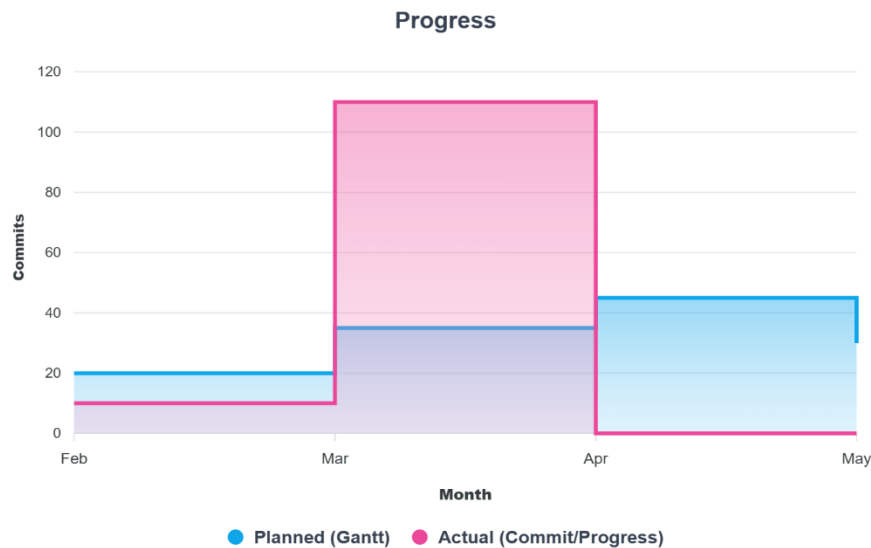


Fig: Area chart reflecting progress on git commits by feature

1.4 Current System Capabilities (Completed):

To provide a clear picture of the project’s current state, the following list outlines the functional capabilities now live in the CogniCare prototype for each user role:

User Authentication & Registration:

- **Role-Specific Registration:** A unified registration flow where users must self-identify as either a **Patient** or a **Carer**. This choice dynamically initializes the correct database schema in **PostgreSQL** via **Prisma**.
- **RBAC-Gated Login:** Secure login handled by **NextAuth.js** that assigns a persistent role-based token (JWT). Upon login, the system automatically routes Patients to the “Mind Dump” interface and Carers to the “Patient Overview” dashboard.
- **Secure Session Management:** Implementation of **Capacitor Secure Storage** to keep authentication tokens encrypted on the mobile device hardware.

Patient-Facing Functionality:

- **Voice-Enabled “Mind Dump”:** Patients can record unstructured audio logs using their phone’s microphone via the integrated **Capacitor Voice Recorder**.
- **Privacy-First Processing:** Integration of **@yellowsakura/js-pii-mask** to redact PII locally before data is transmitted to the AI engine.

Carer-Facing Functionality:

- **Secure Patient Linking:** Carers can only view health data for patients who have granted them explicit permissions, enforced by the backend RBAC middleware.

- **Carer Observation Logic:** Carers can view the patient logs and can add observation note by clicking the “Add Note” button in carer’s log view.

Shared Functionality between carer and patient:

- **Automated Clinical Mapping:** Raw logs are automatically categorized by the **Gemini 2.5 Flash** engine into the “5 Clinical Pillars” (Physical, Mood, Cognitive, Sleep, Social).
- **Clinical PDF Generation:** Capability to generate and open professional multi-page health summaries on-device using **jsPDF** and **html2canvas** for medical consultations.

1.5 Status of Requirements (Functional Implementation for ongoing and pending):

STARTED (Currently in Progress):

- **Visualization Insights:** Currently the application is able to display the data based on just patient’s log. I need to update the report with further categorization of carer and patient logs in pdf view.
- **Carer View Refactoring:** My current functionality is only patient can log symptoms and carer can just add observation note, but it should be both, if patient gives permission, then, carer can log symptoms on behalf of patient.

PENDING (Awaiting Development):

- **Security Audit:** Final verification of PII scrubbing efficiency and penetration testing of the decoupled API endpoints.
- **Real-time Care Circle Messaging:** Backend logic for **Socket.io** is 100% complete; **Frontend UI components** for the chat interface are currently being integrated and tested for mobile responsiveness.
- **Notification UI refactoring:** Need to adjust the notification tab to logs page, so I can display “Care Circle” tab on patient’s view.

2 Revised project plan:

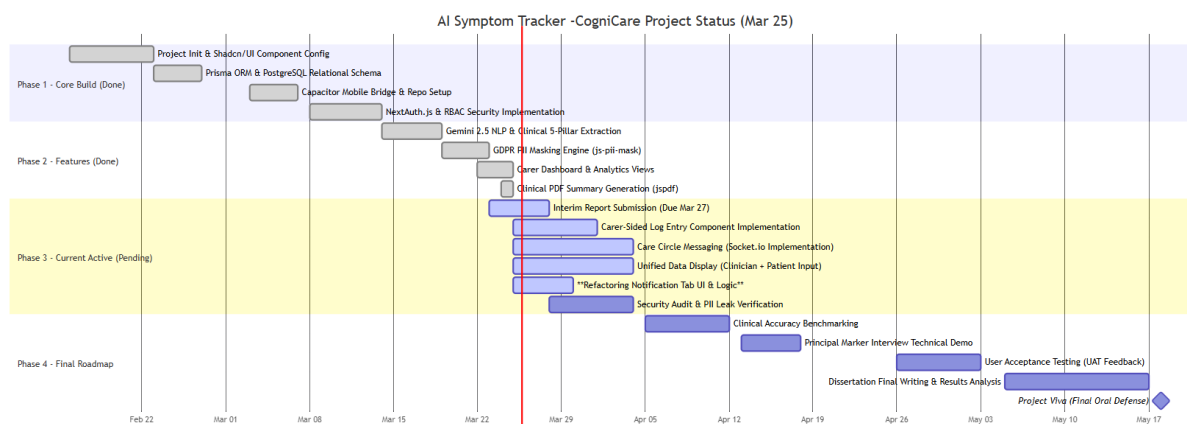


Fig: Gantt chart

📅 As of Mar 25, 2026

Phase 1 — Core Build · Complete				
Task	Start	End	Duration	Status
Project init & Shadcn/UI component config	Feb 16	Feb 22	7d	● Done
Prisma ORM & PostgreSQL relational schema	Feb 23	Feb 26	4d	● Done
Capacitor mobile bridge & repo setup	Mar 3	Mar 6	4d	● Done
NextAuth.js & RBAC security implementation	Mar 8	Mar 13	6d	● Done
Phase 2 — Features · Complete				
Task	Start	End	Duration	Status
Gemini 2.5 NLP & clinical 5-pillar extraction	Mar 14	Mar 18	5d	● Done
GDPR PII masking engine (js-pii-mask)	Mar 19	Mar 22	4d	● Done
Carer dashboard & analytics views	Mar 22	Mar 24	3d	● Done
Clinical PDF summary generation (jspdf)	Mar 24	Mar 24	1d	● Done
Phase 3 — Current Active				
Task	Start	End	Duration	Status
⚠️ Interim report submission	Mar 23	Mar 27	5d	● Active
Carer-sided log entry component	Mar 25	Mar 31	7d	● Active
Care circle messaging (Socket.io)	Mar 25	Apr 3	10d	● Active
Unified data display (clinician + patient input)	Mar 25	Apr 3	10d	● Active
Refactoring notification tab UI & logic	Mar 25	Mar 29	5d	● Active
Security audit & PII leak verification	Mar 28	Apr 3	7d	● Upcoming
Phase 4 — Final Roadmap				
Task	Start	End	Duration	Status
Clinical accuracy benchmarking	Apr 5	Apr 11	7d	● Upcoming
Principal marker interview & technical demo	Apr 13	Apr 17	5d	● Upcoming
User acceptance testing (UAT feedback)	Apr 26	May 2	7d	● Upcoming
Dissertation final writing & results analysis	May 5	May 16	12d	● Upcoming
🎓 Project viva — final oral defence	May 18	May 18	—	● Milestone

Fig: Gantt chart detailed project plan